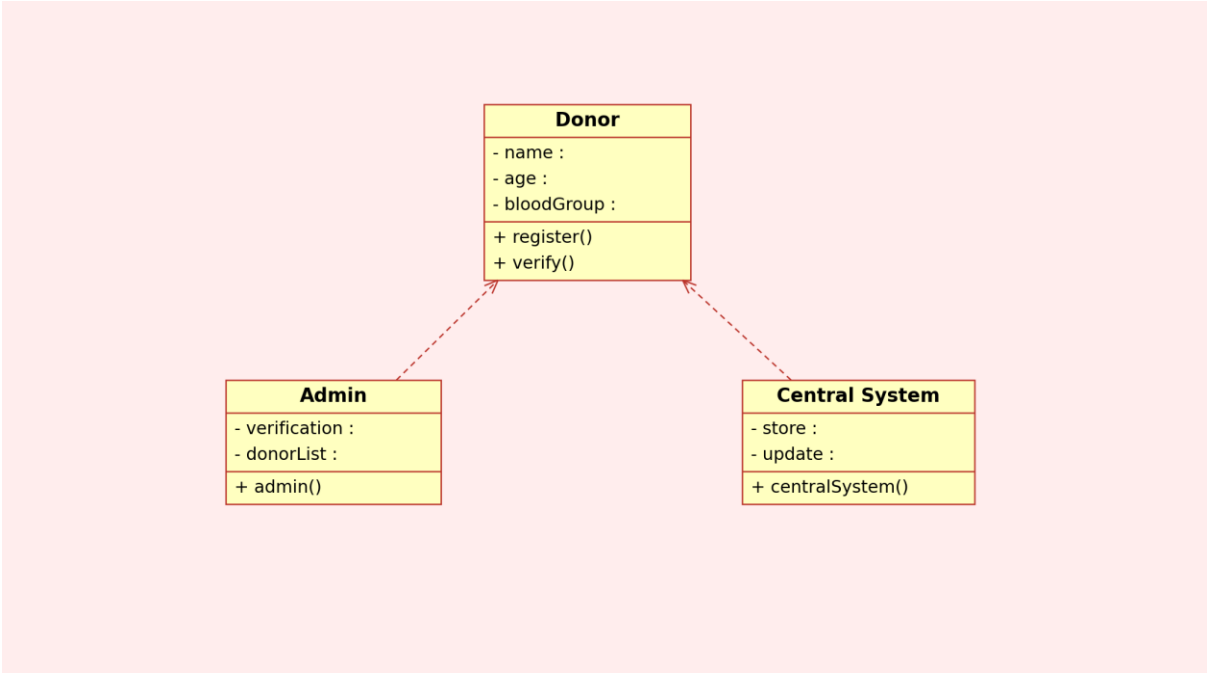
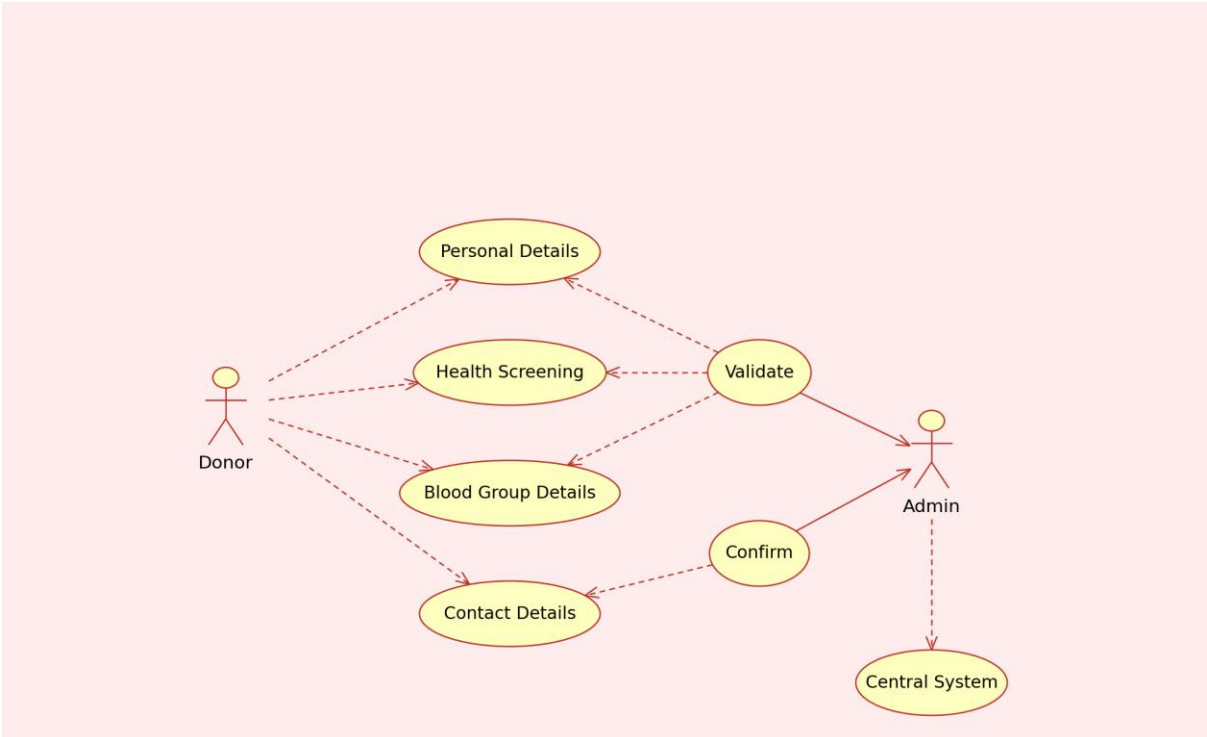
Ex No: 15 – BLOOD BANK REGISTRATION SYSTEM

NAME: HEMALAKSHMI H (192571049)

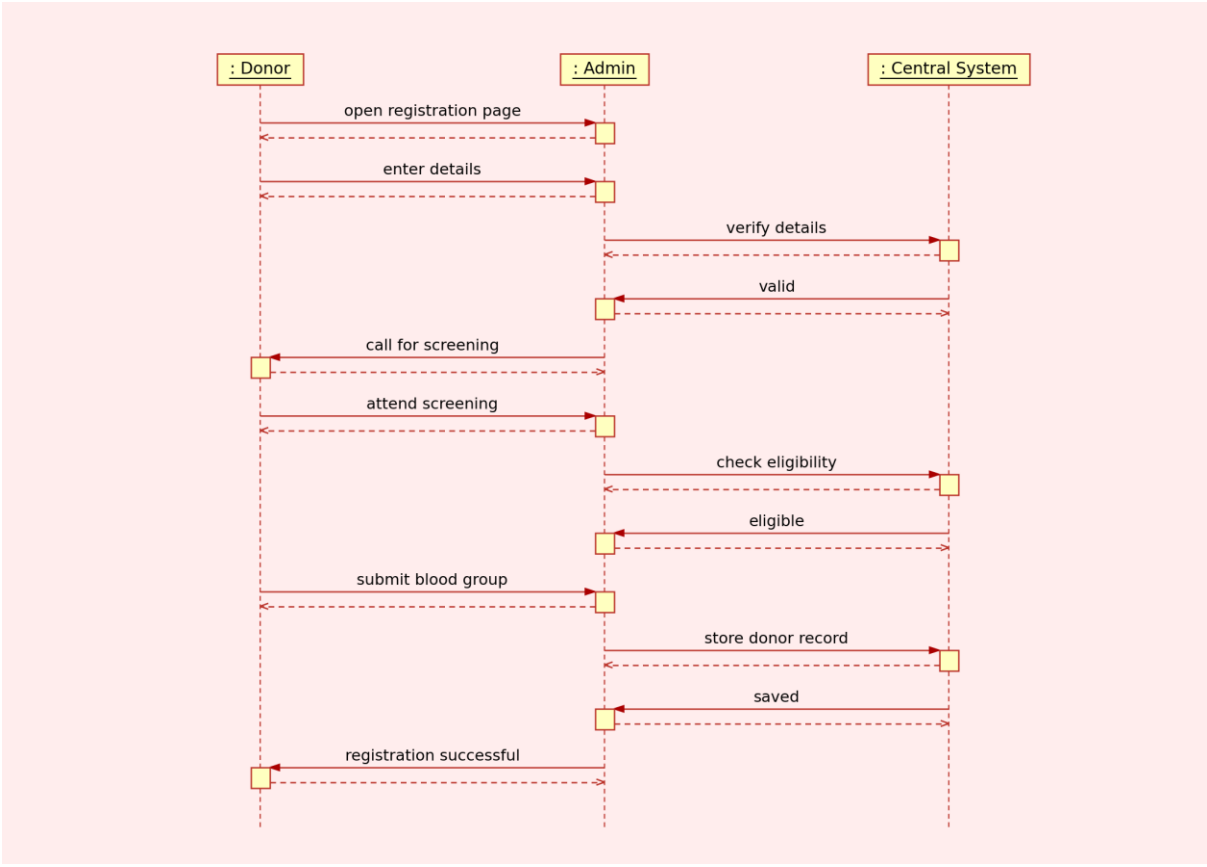
CLASS DIAGRAM:



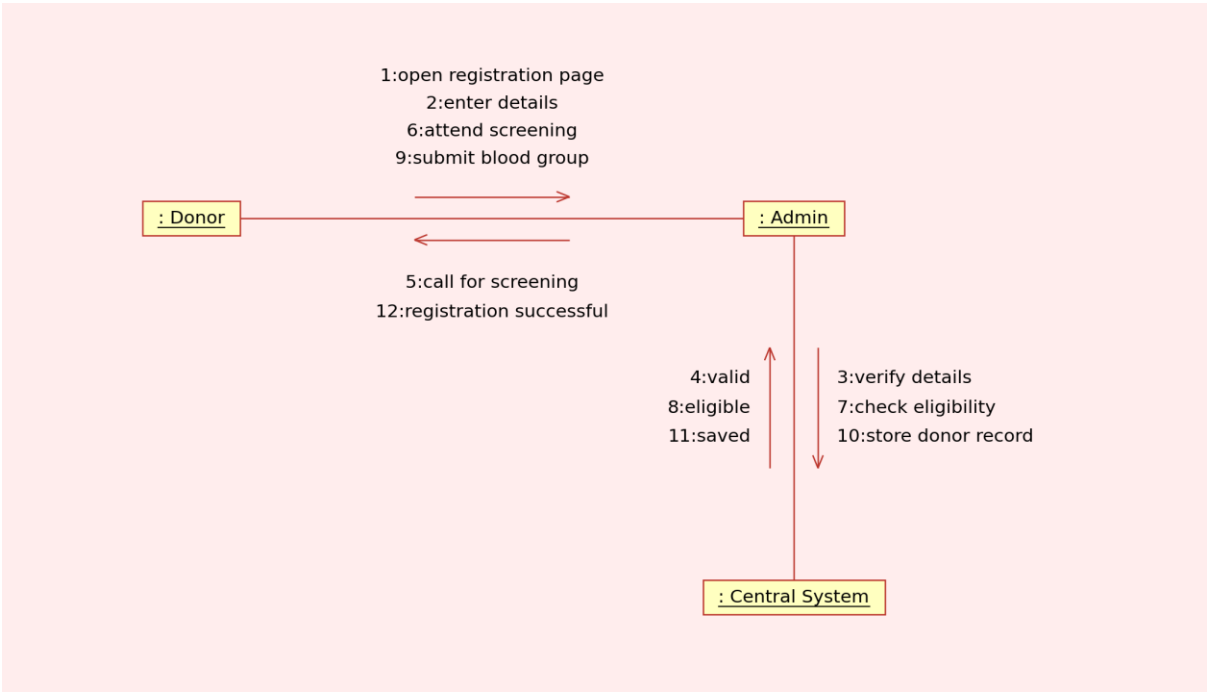
USE CASE DIAGRAM:



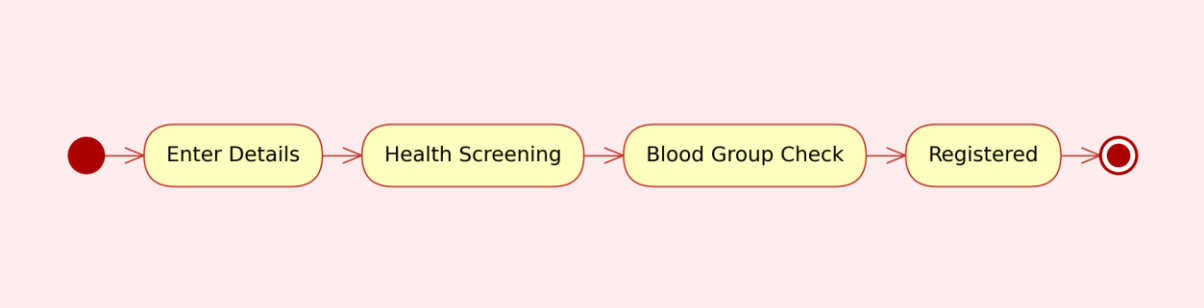
SEQUENCE DIAGRAM:



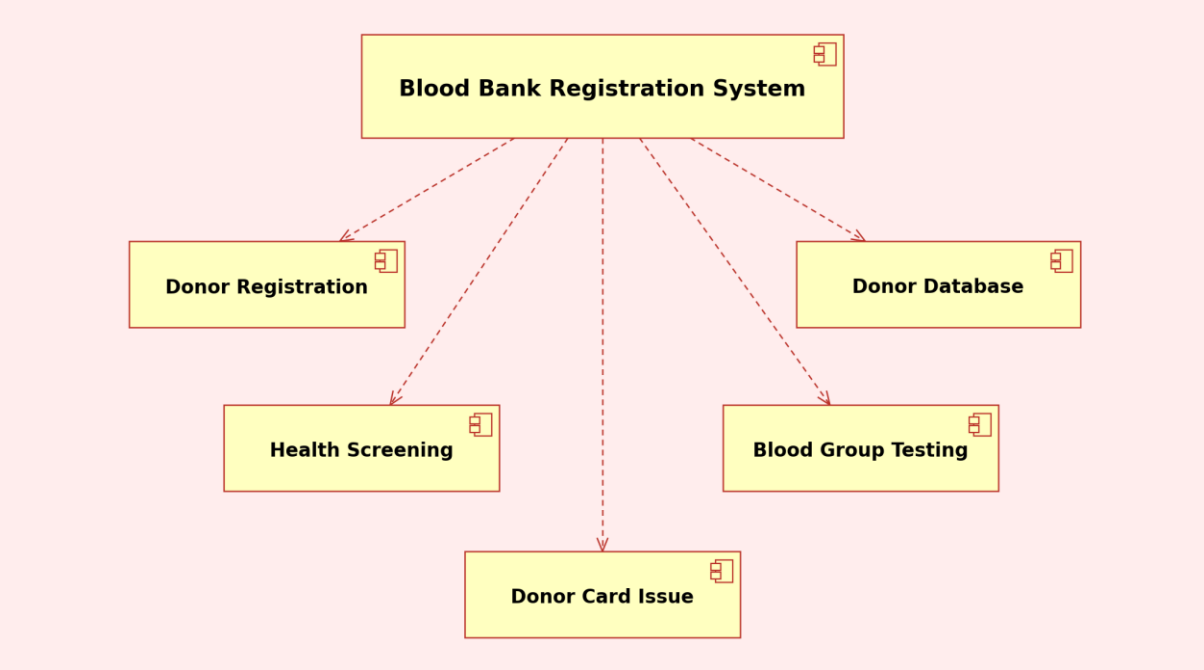
COLLABORATION DIAGRAM:



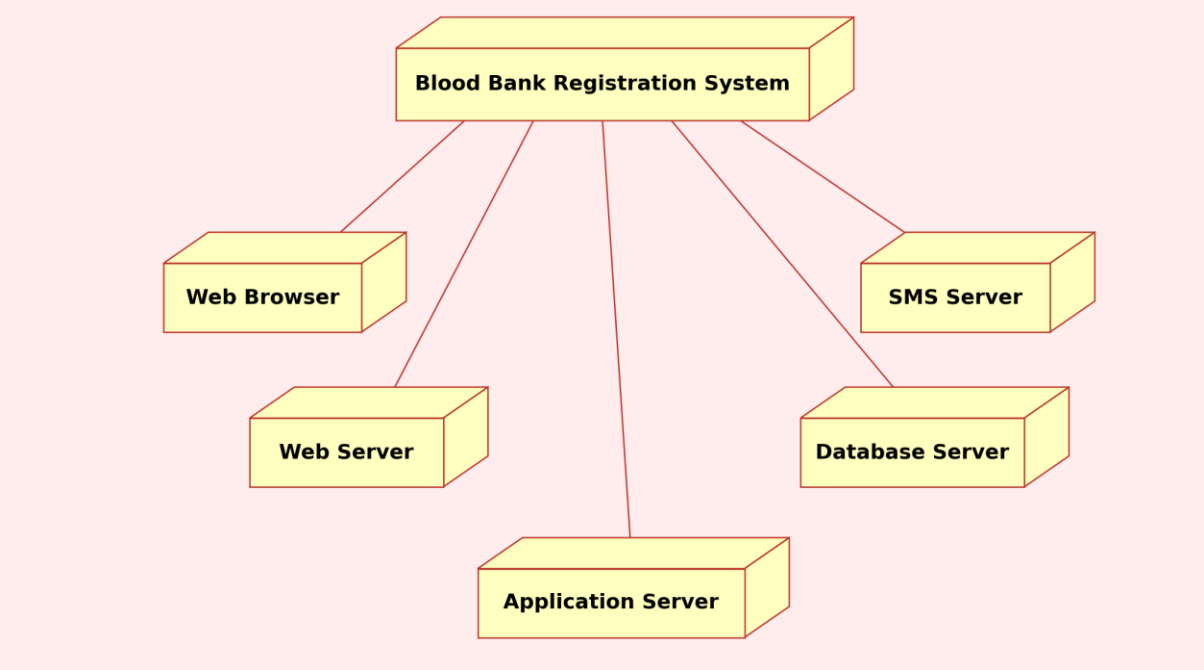
STATE DIAGRAM:



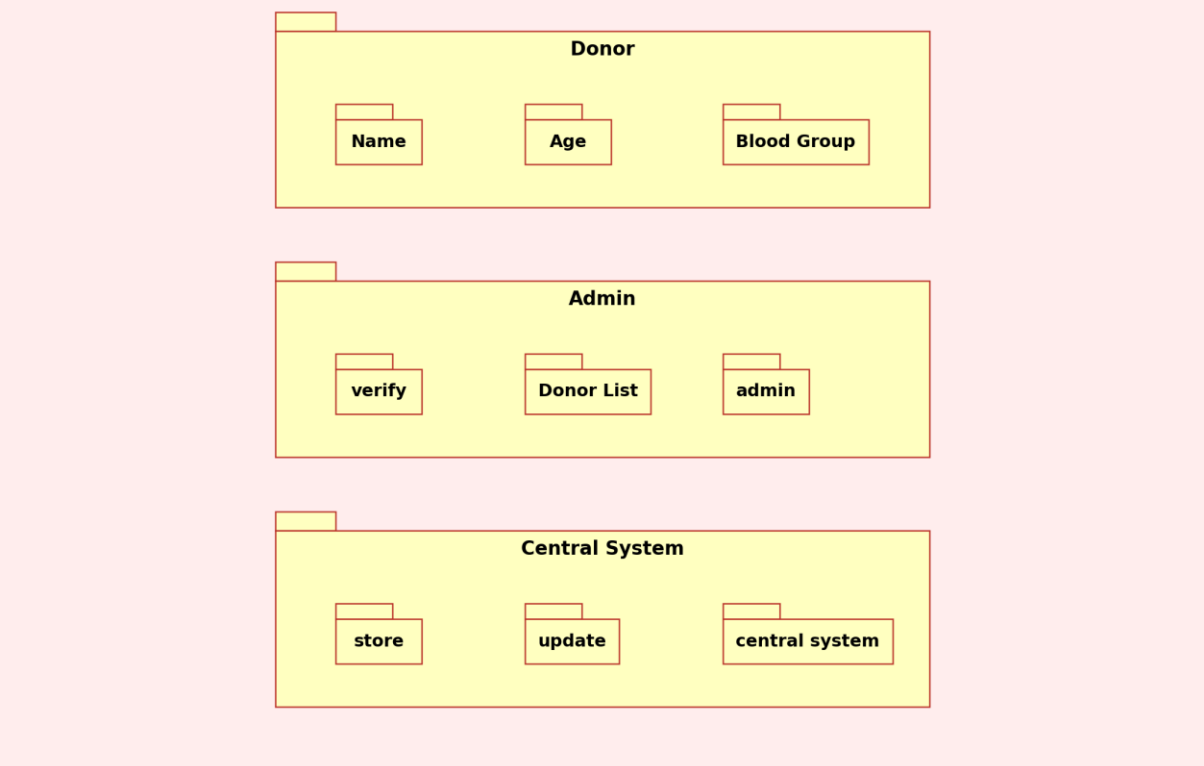
COMPONENT DIAGRAM:



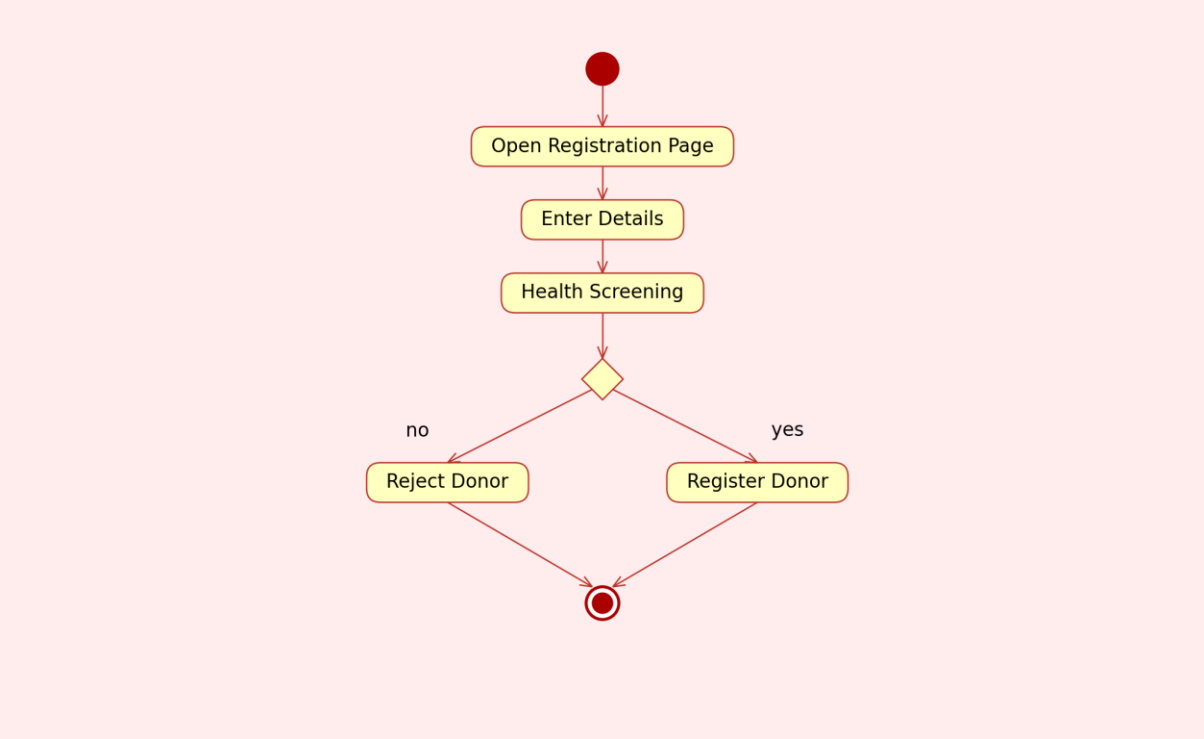
DEPLOYMENT DIAGRAM:



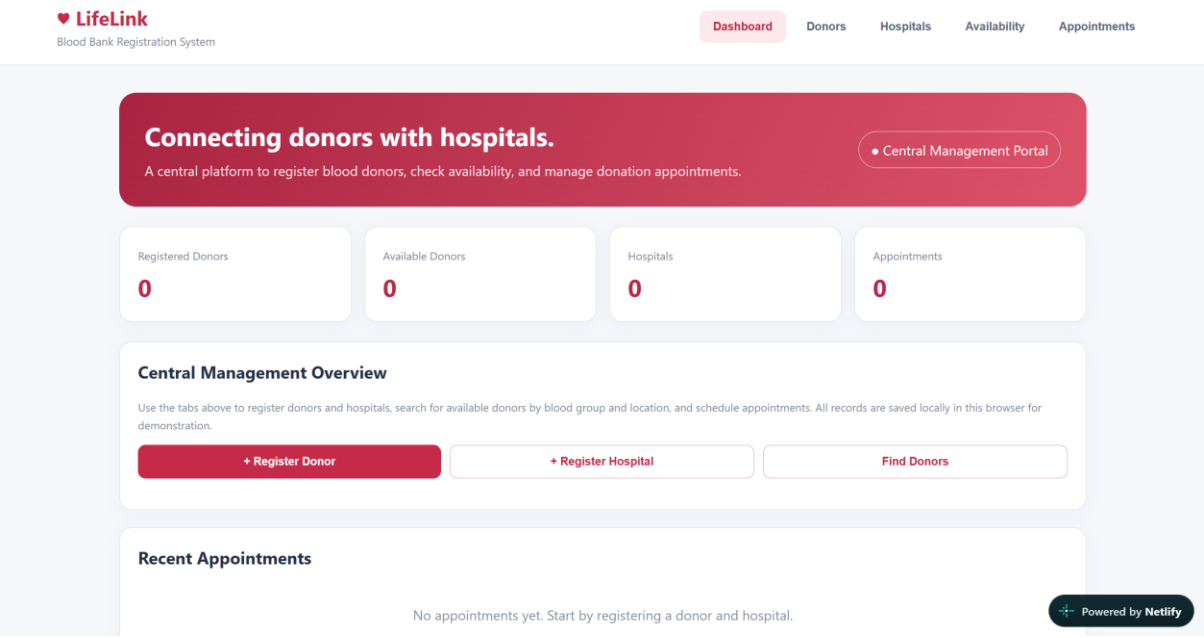
PACKAGE DIAGRAM:

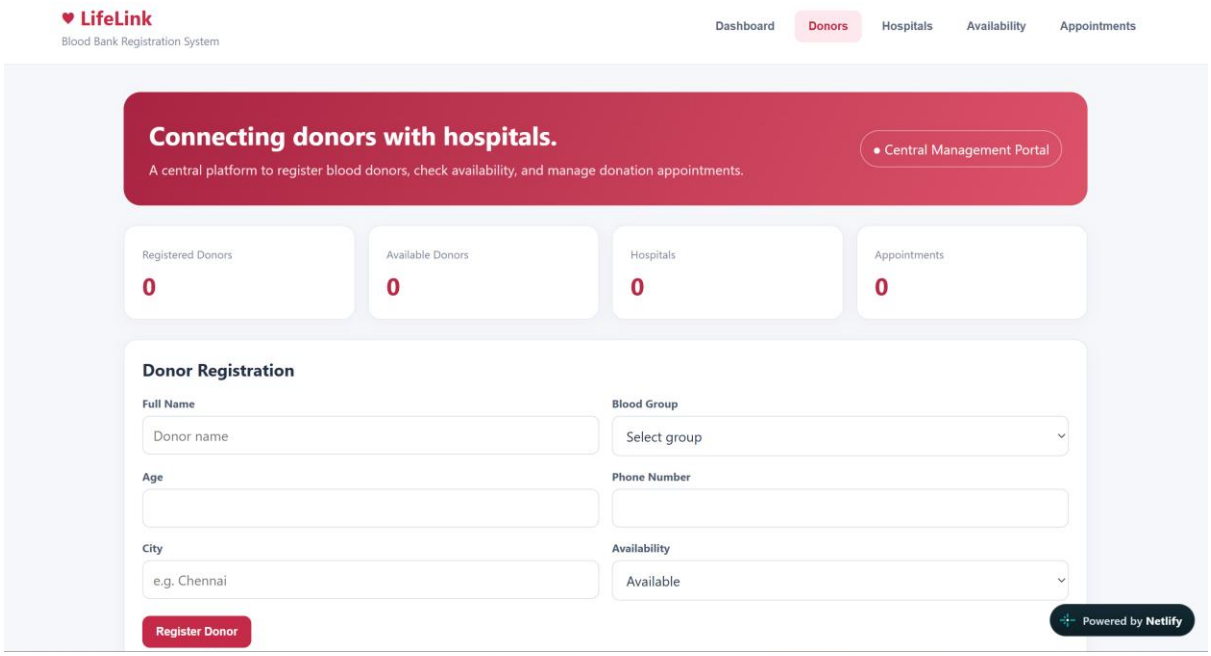


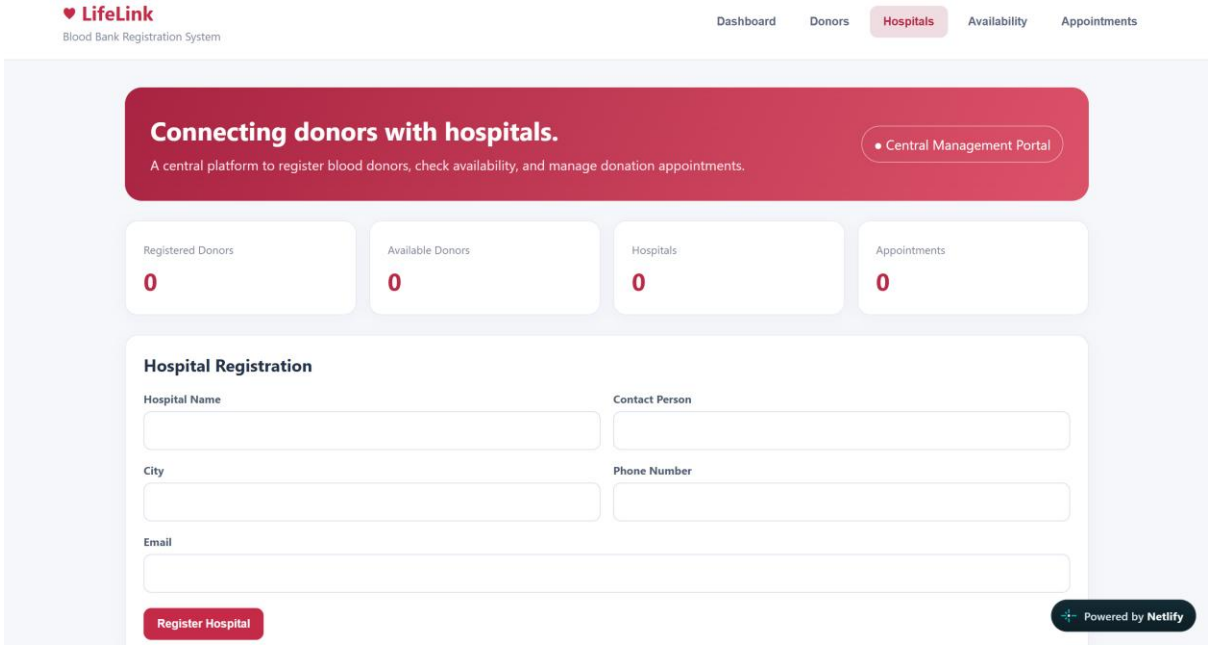
ACTIVITY DIAGRAM:

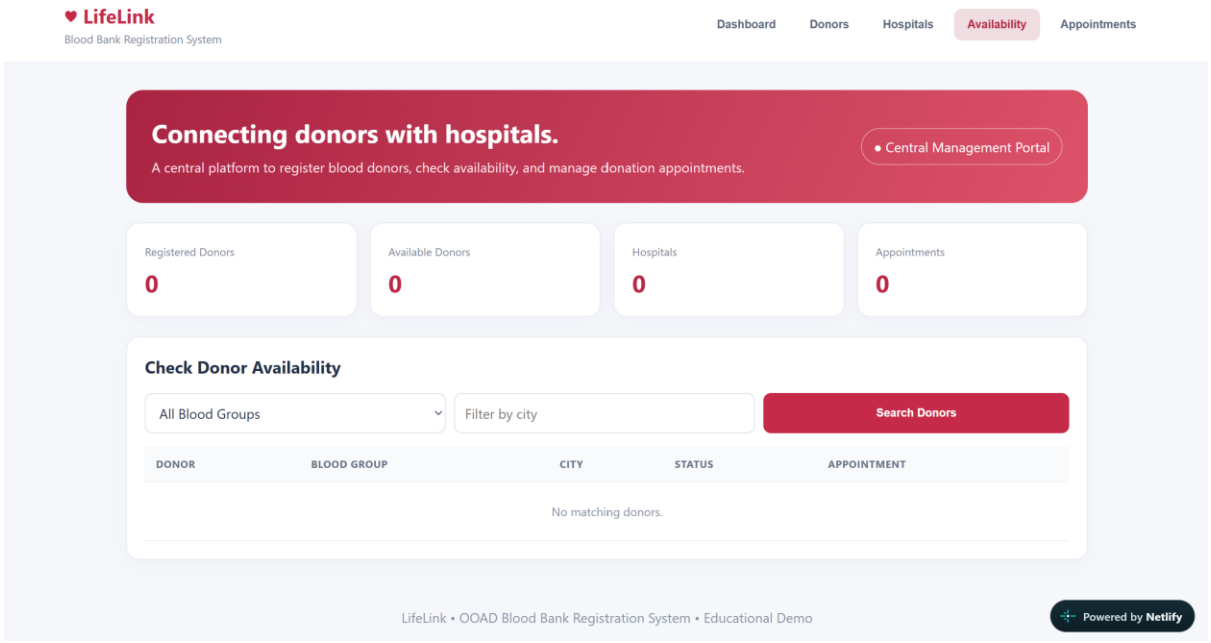


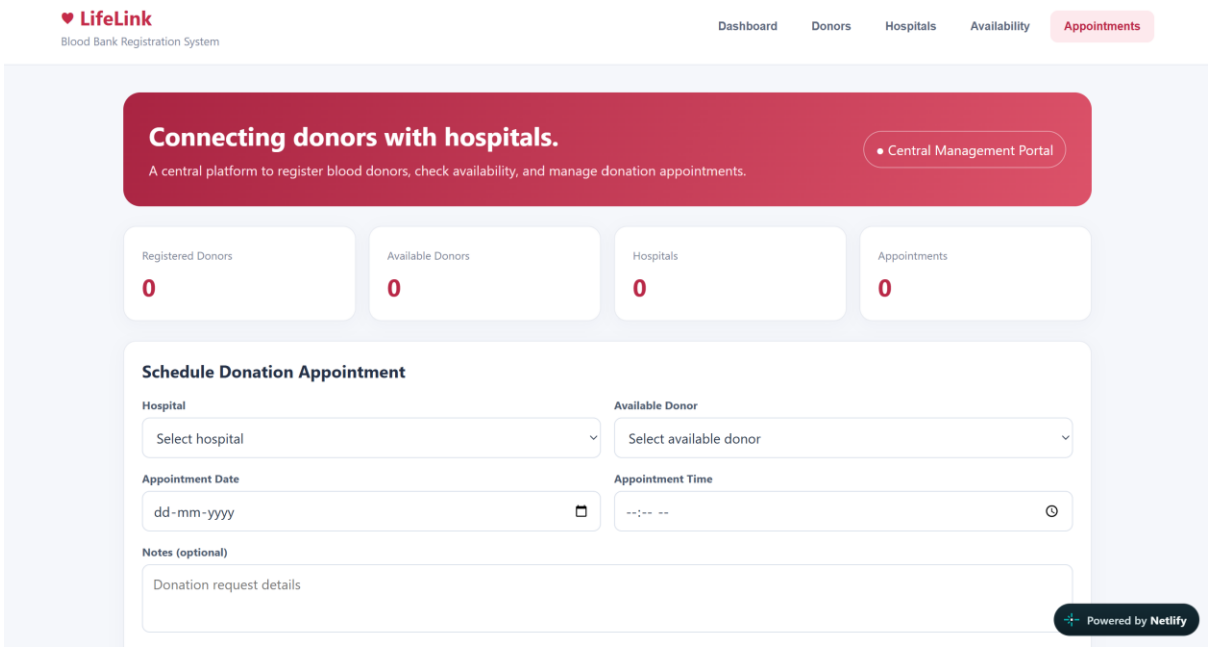
WEPAGE FOR BLOOD BANK REGISTRATION SYSTEM:



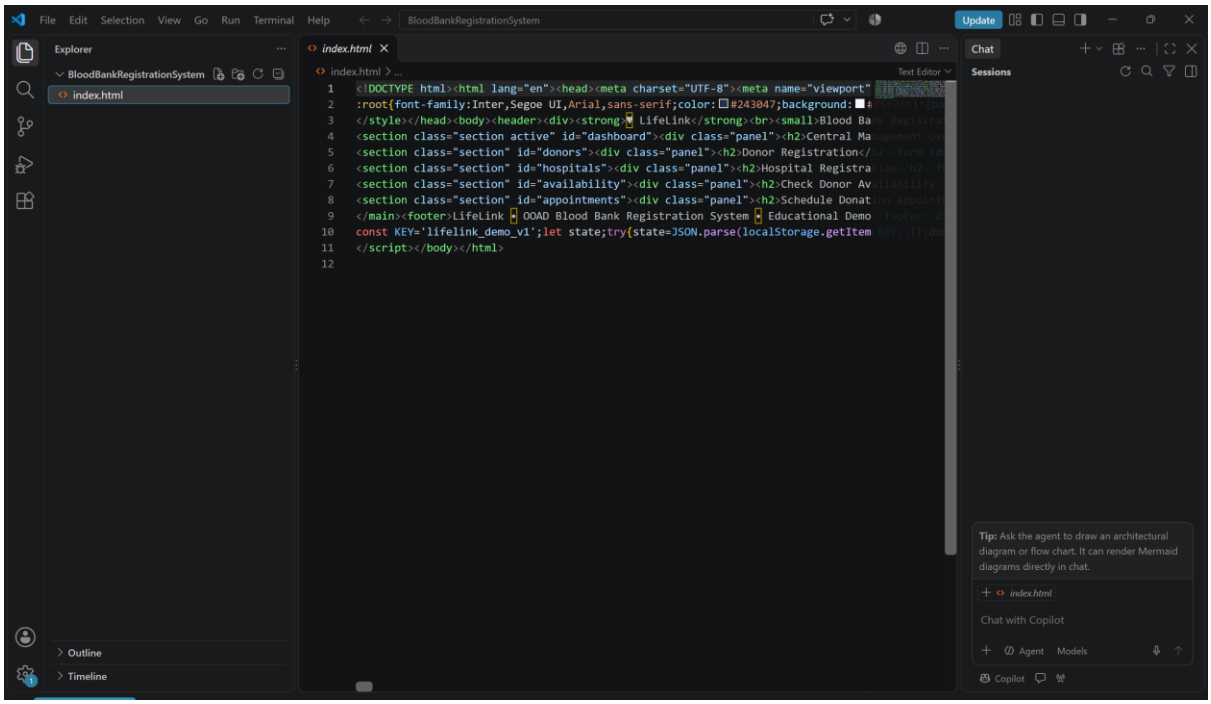

LifeLink
Blood Bank Registration System
Dashboard
Donors
Hospitals
Availability
Appointments
Connecting donors with hospitals.
A central platform to register blood donors, check availability, and manage donation appointments.
• Central Management Portal
Registered Donors
0
Available Donors
0
Hospitals
0
Appointments
0
Donor Registration
Full Name
Donor name
Blood Group
Select group
Age
Phone Number
City
e.g. Chennai
Availability
Available
Register Donor
Powered by Netlify


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0
Available Donors
0
Hospitals
0
Appointments
0
Hospital Registration
Hospital Name
Contact Person
City
Phone Number
Email
Register Hospital
Powered by Netlify

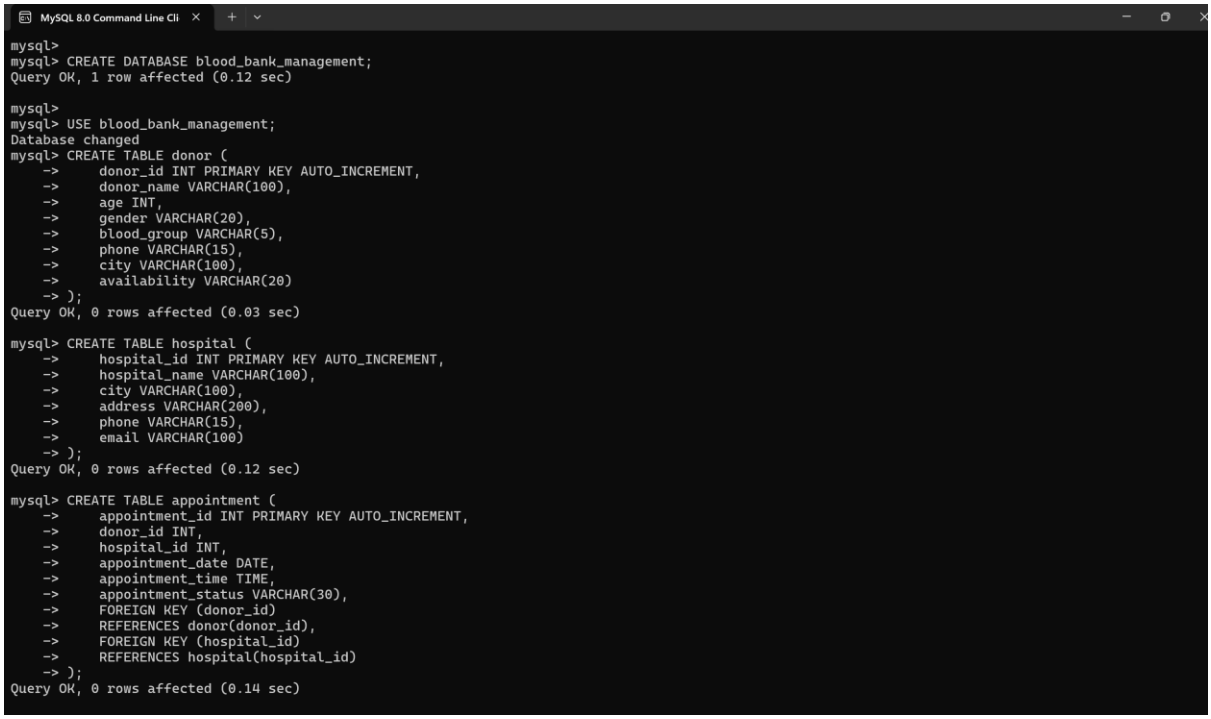

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0
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0
Hospitals
0
Appointments
0
Check Donor Availability
All Blood Groups
Filter by city
Search Donors
DONOR
BLOOD GROUP
CITY
STATUS
APPOINTMENT
No matching donors.
LifeLink • OOAD Blood Bank Registration System • Educational Demo
Powered by Netlify

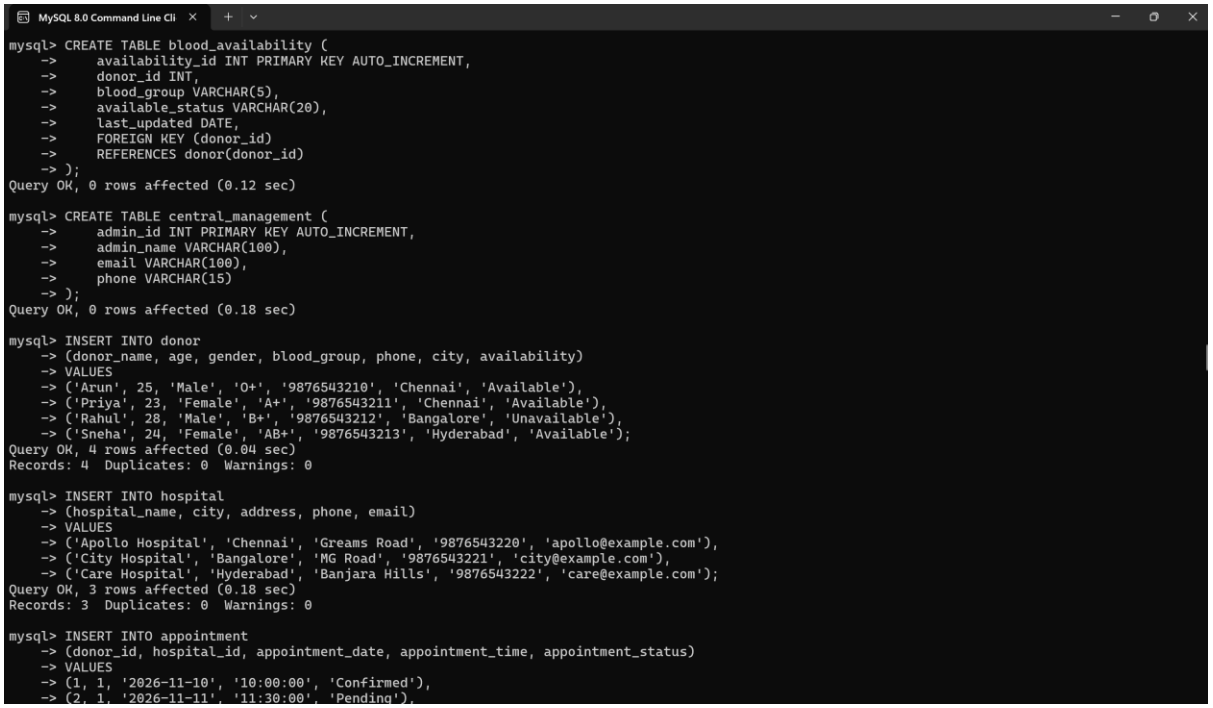

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0
Available Donors
0
Hospitals
0
Appointments
0
Schedule Donation Appointment
Hospital
Select hospital
Available Donor
Select available donor
Appointment Date
dd-mm-yyyy
Appointment Time
--:-- --
Notes (optional)
Donation request details
Powered by Netlify

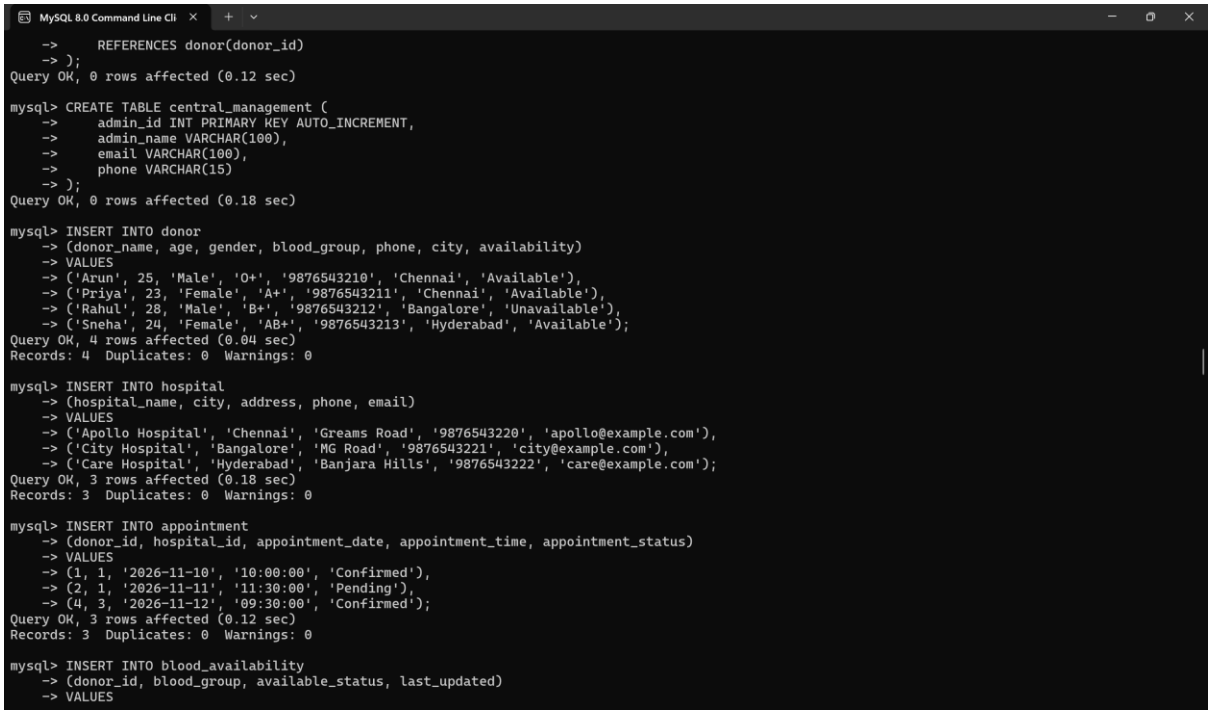
HTML:

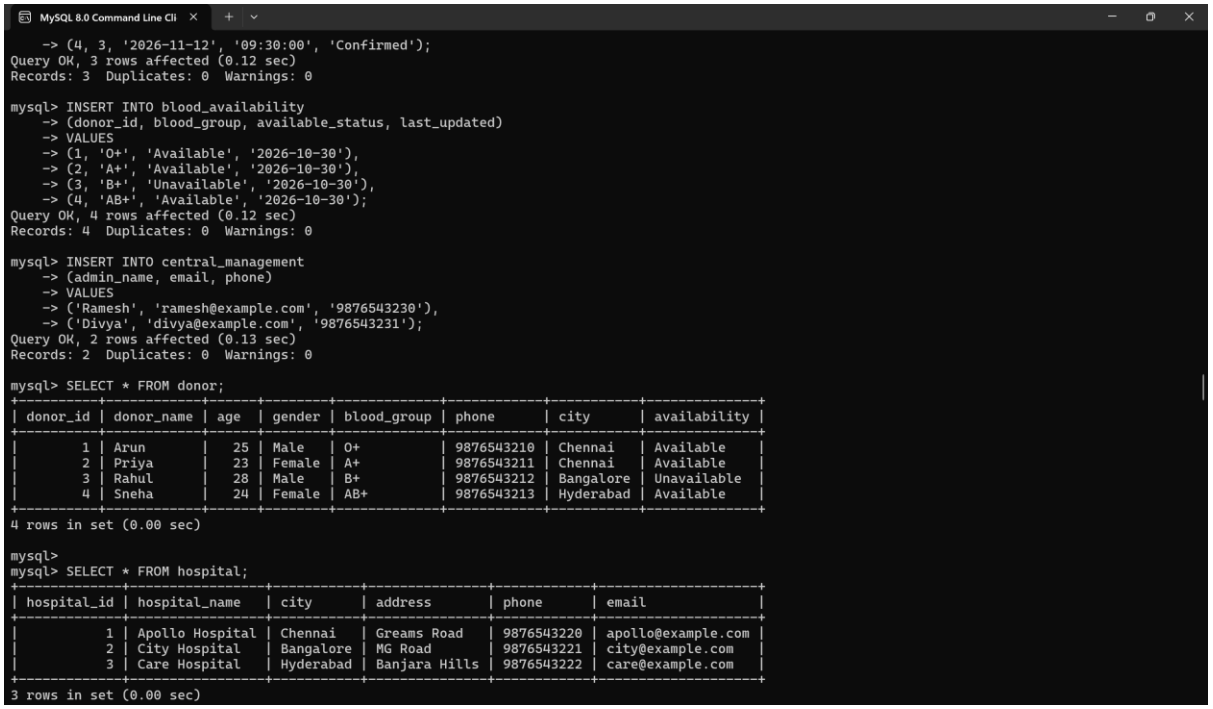


MYSQL DATABASE:




MySQL 8.0 Command Line Cli
mysql> CREATE TABLE blood_availability (
    ->   availability_id INT PRIMARY KEY AUTO_INCREMENT,
    ->   donor_id INT,
    ->   blood_group VARCHAR(5),
    ->   available_status VARCHAR(20),
    ->   last_updated DATE,
    ->   FOREIGN KEY (donor_id)
    ->   REFERENCES donor(donor_id)
    ->);
Query OK, 0 rows affected (0.12 sec)
mysql> CREATE TABLE central_management (
    ->   admin_id INT PRIMARY KEY AUTO_INCREMENT,
    ->   admin_name VARCHAR(100),
    ->   email VARCHAR(100),
    ->   phone VARCHAR(15)
    ->);
Query OK, 0 rows affected (0.18 sec)
mysql> INSERT INTO donor
    -> (donor_name, age, gender, blood_group, phone, city, availability)
    -> VALUES
    -> ('Arun', 25, 'Male', 'O+', '9876543210', 'Chennai', 'Available'),
    -> ('Priya', 23, 'Female', 'A+', '9876543211', 'Chennai', 'Available'),
    -> ('Rahul', 28, 'Male', 'B+', '9876543212', 'Bangalore', 'Unavailable'),
    -> ('Sneha', 24, 'Female', 'AB+', '9876543213', 'Hyderabad', 'Available');
Query OK, 4 rows affected (0.04 sec)
Records: 4 Duplicates: 0 Warnings: 0
mysql> INSERT INTO hospital
    -> (hospital_name, city, address, phone, email)
    -> VALUES
    -> ('Apollo Hospital', 'Chennai', 'Greams Road', '9876543220', 'apollo@example.com'),
    -> ('City Hospital', 'Bangalore', 'MG Road', '9876543221', 'city@example.com'),
    -> ('Care Hospital', 'Hyderabad', 'Banjara Hills', '9876543222', 'care@example.com');
Query OK, 3 rows affected (0.18 sec)
Records: 3 Duplicates: 0 Warnings: 0
mysql> INSERT INTO appointment
    -> (donor_id, hospital_id, appointment_date, appointment_time, appointment_status)
    -> VALUES
    -> (1, 1, '2026-11-10', '10:00:00', 'Confirmed'),
    -> (2, 1, '2026-11-11', '11:30:00', 'Pending'),


MySQL 8.0 Command Line Cli
    ->   REFERENCES donor(donor_id)
    ->);
Query OK, 0 rows affected (0.12 sec)
mysql> CREATE TABLE central_management (
    ->   admin_id INT PRIMARY KEY AUTO_INCREMENT,
    ->   admin_name VARCHAR(100),
    ->   email VARCHAR(100),
    ->   phone VARCHAR(15)
    ->);
Query OK, 0 rows affected (0.18 sec)
mysql> INSERT INTO donor
    -> (donor_name, age, gender, blood_group, phone, city, availability)
    -> VALUES
    -> ('Arun', 25, 'Male', 'O+', '9876543210', 'Chennai', 'Available'),
    -> ('Priya', 23, 'Female', 'A+', '9876543211', 'Chennai', 'Available'),
    -> ('Rahul', 28, 'Male', 'B+', '9876543212', 'Bangalore', 'Unavailable'),
    -> ('Sneha', 24, 'Female', 'AB+', '9876543213', 'Hyderabad', 'Available');
Query OK, 4 rows affected (0.04 sec)
Records: 4 Duplicates: 0 Warnings: 0
mysql> INSERT INTO hospital
    -> (hospital_name, city, address, phone, email)
    -> VALUES
    -> ('Apollo Hospital', 'Chennai', 'Greams Road', '9876543220', 'apollo@example.com'),
    -> ('City Hospital', 'Bangalore', 'MG Road', '9876543221', 'city@example.com'),
    -> ('Care Hospital', 'Hyderabad', 'Banjara Hills', '9876543222', 'care@example.com');
Query OK, 3 rows affected (0.18 sec)
Records: 3 Duplicates: 0 Warnings: 0
mysql> INSERT INTO appointment
    -> (donor_id, hospital_id, appointment_date, appointment_time, appointment_status)
    -> VALUES
    -> (1, 1, '2026-11-10', '10:00:00', 'Confirmed'),
    -> (2, 1, '2026-11-11', '11:30:00', 'Pending'),
    -> (4, 3, '2026-11-12', '09:30:00', 'Confirmed');
Query OK, 3 rows affected (0.12 sec)
Records: 3 Duplicates: 0 Warnings: 0
mysql> INSERT INTO blood_availability
    -> (donor_id, blood_group, available_status, last_updated)
    -> VALUES



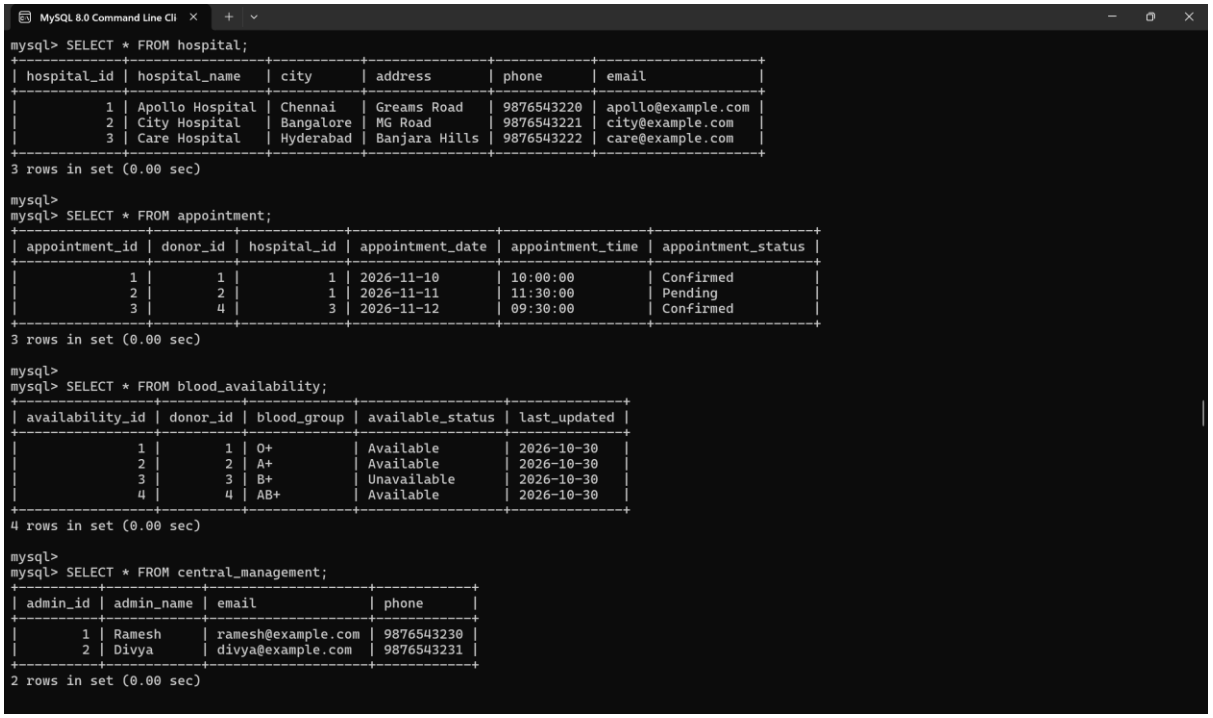

```
    -> (4, 3, '2026-11-12', '09:30:00', 'Confirmed');
Query OK, 3 rows affected (0.12 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> INSERT INTO blood_availability
    -> (donor_id, blood_group, available_status, last_updated)
    -> VALUES
    -> (1, 'O+', 'Available', '2026-10-30'),
    -> (2, 'A+', 'Available', '2026-10-30'),
    -> (3, 'B+', 'Unavailable', '2026-10-30'),
    -> (4, 'AB+', 'Available', '2026-10-30');
Query OK, 4 rows affected (0.12 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> INSERT INTO central_management
    -> (admin_name, email, phone)
    -> VALUES
    -> ('Ramesh', 'ramesh@example.com', '9876543230'),
    -> ('Divya', 'divya@example.com', '9876543231');
Query OK, 2 rows affected (0.13 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM donor;
+----------+------------+------+--------+-------------+------------+-----------+--------------+
| donor_id | donor_name | age  | gender | blood_group | phone      | city      | availability |
+----------+------------+------+--------+-------------+------------+-----------+--------------+
|        1 | Arun       |   25 | Male   | O+          | 9876543210 | Chennai   | Available    |
|        2 | Priya      |   23 | Female | A+          | 9876543211 | Chennai   | Available    |
|        3 | Rahul      |   28 | Male   | B+          | 9876543212 | Bangalore | Unavailable  |
|        4 | Sneha      |   24 | Female | AB+         | 9876543213 | Hyderabad | Available    |
+----------+------------+------+--------+-------------+------------+-----------+--------------+
4 rows in set (0.00 sec)

mysql>
mysql> SELECT * FROM hospital;
+-------------+----------------+-----------+---------------+------------+--------------------+
| hospital_id | hospital_name  | city      | address       | phone      | email              |
+-------------+----------------+-----------+---------------+------------+--------------------+
|           1 | Apollo Hospital | Chennai   | Greams Road   | 9876543220 | apollo@example.com |
|           2 | City Hospital   | Bangalore | MG Road       | 9876543221 | city@example.com   |
|           3 | Care Hospital   | Hyderabad | Banjara Hills | 9876543222 | care@example.com   |
+-------------+----------------+-----------+---------------+------------+--------------------+
3 rows in set (0.00 sec)
```

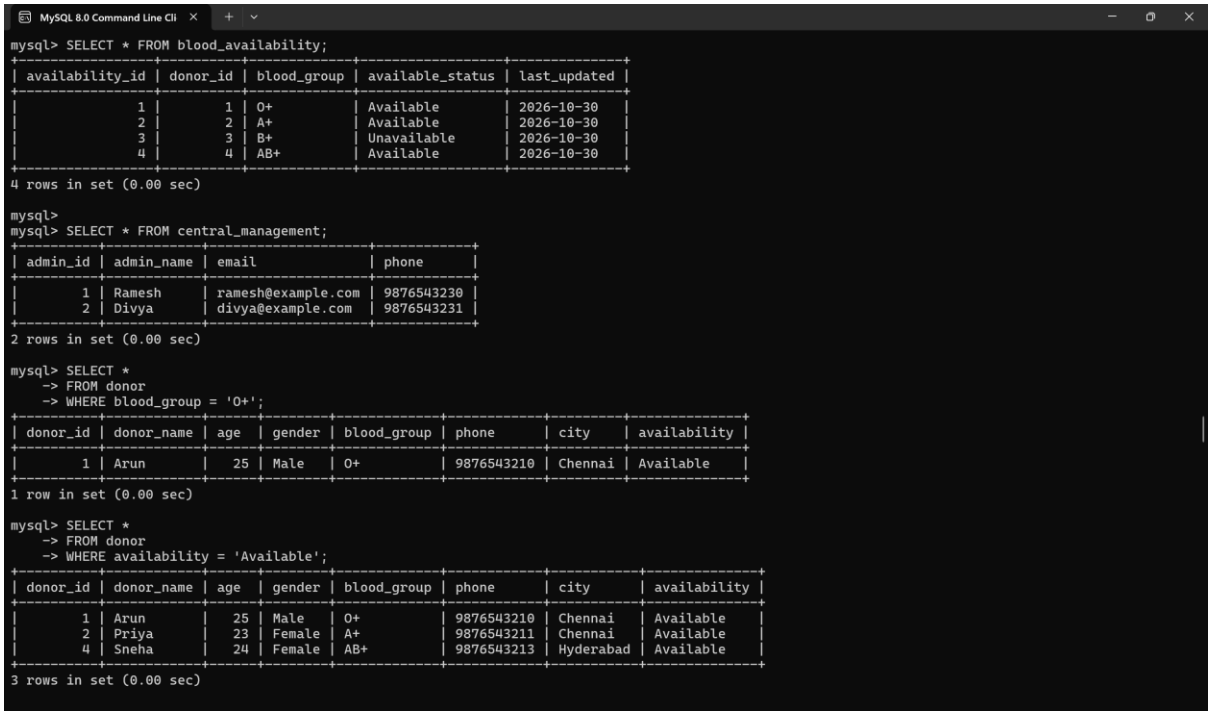



```
mysql> SELECT * FROM hospital;
+-------------+----------------+-----------+---------------+------------+--------------------+
| hospital_id | hospital_name  | city      | address       | phone      | email              |
+-------------+----------------+-----------+---------------+------------+--------------------+
|           1 | Apollo Hospital | Chennai   | Greams Road   | 9876543220 | apollo@example.com |
|           2 | City Hospital   | Bangalore | MG Road       | 9876543221 | city@example.com   |
|           3 | Care Hospital   | Hyderabad | Banjara Hills | 9876543222 | care@example.com   |
+-------------+----------------+-----------+---------------+------------+--------------------+
3 rows in set (0.00 sec)

mysql>
mysql> SELECT * FROM appointment;
+----------------+----------+-------------+------------------+------------------+--------------------+
| appointment_id | donor_id | hospital_id | appointment_date | appointment_time | appointment_status |
+----------------+----------+-------------+------------------+------------------+--------------------+
|              1 |        1 |           1 | 2026-11-10       | 10:00:00         | Confirmed          |
|              2 |        2 |           1 | 2026-11-11       | 11:30:00         | Pending            |
|              3 |        4 |           3 | 2026-11-12       | 09:30:00         | Confirmed          |
+----------------+----------+-------------+------------------+------------------+--------------------+
3 rows in set (0.00 sec)

mysql>
mysql> SELECT * FROM blood_availability;
+-----------------+----------+-------------+------------------+--------------+
| availability_id | donor_id | blood_group | available_status | last_updated |
+-----------------+----------+-------------+------------------+--------------+
|               1 |        1 | O+          | Available        | 2026-10-30   |
|               2 |        2 | A+          | Available        | 2026-10-30   |
|               3 |        3 | B+          | Unavailable      | 2026-10-30   |
|               4 |        4 | AB+         | Available        | 2026-10-30   |
+-----------------+----------+-------------+------------------+--------------+
4 rows in set (0.00 sec)

mysql>
mysql> SELECT * FROM central_management;
+----------+------------+--------------------+------------+
| admin_id | admin_name | email              | phone      |
+----------+------------+--------------------+------------+
|        1 | Ramesh     | ramesh@example.com | 9876543230 |
|        2 | Divya      | divya@example.com  | 9876543231 |
+----------+------------+--------------------+------------+
2 rows in set (0.00 sec)
```


MySQL 8.0 Command Line Cli
mysql> SELECT * FROM blood_availability;
+-----------------+----------+-------------+-------------------+--------------+
| availability_id | donor_id | blood_group | available_status | last_updated |
+-----------------+----------+-------------+-------------------+--------------+
1	1	O+	Available	2026-10-30
2	2	A+	Available	2026-10-30
3	3	B+	Unavailable	2026-10-30
4	4	AB+	Available	2026-10-30
+-----------------+----------+-------------+-------------------+--------------+
4 rows in set (0.00 sec)
mysql>
mysql> SELECT * FROM central_management;
+----------+------------+--------------------+------------+
| admin_id | admin_name | email | phone |
+----------+------------+--------------------+------------+
| 1 | Ramesh | ramesh@example.com | 9876543230 |
| 2 | Divya | divya@example.com | 9876543231 |
+----------+------------+--------------------+------------+
2 rows in set (0.00 sec)
mysql> SELECT *
-> FROM donor
-> WHERE blood_group = 'O+';
+----------+------------+------+--------+-------------+------------+---------+--------------+
| donor_id | donor_name | age | gender | blood_group | phone | city | availability |
+----------+------------+------+--------+-------------+------------+---------+--------------+
| 1 | Arun | 25 | Male | O+ | 9876543210 | Chennai | Available |
+----------+------------+------+--------+-------------+------------+---------+--------------+
1 row in set (0.00 sec)
mysql> SELECT *
-> FROM donor
-> WHERE availability = 'Available';
+----------+------------+------+--------+-------------+------------+-----------+--------------+
| donor_id | donor_name | age | gender | blood_group | phone | city | availability |
+----------+------------+------+--------+-------------+------------+-----------+--------------+
1	Arun	25	Male	O+	9876543210	Chennai	Available
2	Priya	23	Female	A+	9876543211	Chennai	Available
4	Sneha	24	Female	AB+	9876543213	Hyderabad	Available
+----------+------------+------+--------+-------------+------------+-----------+--------------+
3 rows in set (0.00 sec)